BITS, TRITS AND CYCLOTOMIC INTEGERS ON ONE PAGE

Two colours on a square, three on a hexagon

A bit is a remainder in the Gaussian integers and a trit is a remainder in the Eisenstein integers.

THE RULE

Divide by the prime over 2 or over 3

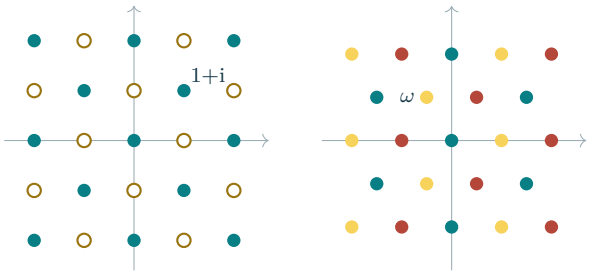
In $\mathbb{Z}[\mathrm{i}]$ the prime $1+\mathrm{i}$ divides 2, and in $\mathbb{Z}[\omega]$ the prime $1-\omega$ divides 3; each leaves exactly as many remainders as its norm, and those remainders are the digits:

$$a+b\,\mathrm{i} \equiv a+b \pmod 2, \qquad a+b\,\omega \equiv a+b \pmod 3$$
$$\mathbb{Z}[\mathrm{i}]/(1+\mathrm{i}) \;\cong\; \{0,1\} \qquad \mathbb{Z}[\omega]/(1-\omega) \;\cong\; \{0,1,2\}$$

The congruences come from $\mathrm{i} \equiv 1$ and $\omega \equiv 1$ modulo those primes. Both primes are special because they meet their rational prime twice,

$$2 = -\mathrm{i}\,(1+\mathrm{i})^2, \qquad 3 = -\omega^2(1-\omega)^2,$$

and 2 and 3 are the only primes that do this in their respective rings.



$\mathbb{Z}[\mathrm{i}]$: two classes $\qquad$ $\mathbb{Z}[\omega]$: three classes

Colouring by remainder gives a checkerboard on the square lattice and a three-colouring on the hexagonal one.

THE FOURTH AND THIRD CYCLOTOMIC FIELDS

Adjoin a root of unity and take its whole numbers

Each field adjoins a primitive root of unity to $\mathbb{Q}$, and its ring of integers is the lattice drawn above; both fields have degree $\varphi(4) = \varphi(3) = 2$, because each root satisfies a quadratic.

	fourth, the Gaussian side	third, the Eisenstein side
root and polynomial	$\mathrm{i} = \mathrm{e}^{2\pi\mathrm{i}/4}$, $\quad \Phi_4(x) = x^2+1$	$\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$, $\quad \Phi_3(x) = x^2+x+1$
field and integers	$\mathbb{Q}(\mathrm{i})$, $\quad \mathbb{Z}[\mathrm{i}] = \{a+b\mathrm{i}\}$	$\mathbb{Q}(\omega)$, $\quad \mathbb{Z}[\omega] = \{a+b\omega\}$
norm	$N(a+b\mathrm{i}) = a^2+b^2$	$N(a+b\omega) = a^2-ab+b^2$
units	$\pm 1, \pm\mathrm{i}$: the four 4th roots	$\pm 1, \pm\omega, \pm\omega^2$: the six 6th roots
digit prime	$1+\mathrm{i}$, norm 2, gives a **bit**	$1-\omega$, norm 3, gives a **trit**

BINARY AND TERNARY INSIDE THE RINGS

A base whose norm is the number of digits

Repeatedly taking the remainder and dividing by the base, exactly as for ordinary numerals, writes every Gaussian integer in base $-1+\mathrm{i}$ with the digits $\{0,1\}$ and every Eisenstein integer in base $\omega-1$ with the digits $\{0,1,2\}$, each expansion finite and unique. Both bases are associates of the digit primes, and their squares are $(-1+\mathrm{i})^2 = -2\mathrm{i}$ and $(\omega-1)^2 = -3\omega$.

binary in base $-1+\mathrm{i}$

$$\mathrm{i} = 11, \qquad 2 = 1100, \qquad -1 = 11101$$

ternary in base $\omega-1$

$$\omega = 11, \qquad 3 = 1200, \qquad -1 = 12102$$

Negative and complex numbers need no sign or second coordinate, since the base itself points in a new direction at every power; checking $3 = (\omega-1)^3 + 2(\omega-1)^2 = (3+6\omega) - 6\omega$ is a good first exercise.

Digits as roots of unity

Sending a bit b to $(-1)^b$ and a trit t to ω^t turns adding digits into multiplying roots of unity, since $(-1)^{b+b'} = (-1)^b(-1)^{b'}$ and $\omega^{t+t'} = \omega^t\omega^{t'}$ with the exponents read modulo 2 and 3.

The same phases in qubits and qutrits

The phase gate on a qubit sends $|b\rangle$ to $(-1)^b|b\rangle$, and the clock gate on a qutrit sends $|t\rangle$ to $\omega^t|t\rangle$, so its phases live in $\mathbb{Z}[\omega]$; the qutrit Fourier transform is built from powers of ω just as the Hadamard is built from ± 1.

Conventions. Throughout, $\omega = \mathrm{e}^{2\pi\mathrm{i}/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$, so $\omega^2+\omega+1 = 0$, and Φ_n is the nth cyclotomic polynomial, whose roots are the primitive nth roots of unity; "degree" means the dimension of the field over $\mathbb{Q}$. Two integers are associates when they differ by a unit factor, as $1-\omega$ and $\omega-1$ do.

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